

(a) Anatomical and developmental readouts

| Model    | Joint AMI $\uparrow$        |                             |                             | Brain DPT $\rho$ $\uparrow$ |              |              | Brain MAE (days) $\downarrow$ |              |              |
|----------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--------------|--------------|-------------------------------|--------------|--------------|
|          | 8                           | 32                          | 128                         | 8                           | 32           | 128          | 8                             | 32           | 128          |
| ZoomST   | <b>0.585</b><br>$\pm 0.006$ | <b>0.571</b><br>$\pm 0.003$ | 0.538<br>$\pm 0.002$        | 0.868                       | <b>0.822</b> | <b>0.354</b> | <b>0.803</b>                  | <b>0.715</b> | <b>0.734</b> |
| CS       | 0.575<br>$\pm 0.003$        | 0.564<br>$\pm 0.001$        | <b>0.539</b><br>$\pm 0.009$ | 0.857                       | 0.737        | 0.143        | 0.818                         | 0.733        | 0.802        |
| No Cross | 0.570<br>$\pm 0.001$        | 0.562<br>$\pm 0.005$        | 0.539<br>$\pm 0.0003$       | <b>0.903</b>                | 0.738        | -0.239       | 0.936                         | 0.853        | 0.929        |

(b) Cross-granularity CKA

| Model    | 8 $\leftrightarrow$ 32 $\uparrow$ | 8 $\leftrightarrow$ 128 $\uparrow$ | 32 $\leftrightarrow$ 128 $\uparrow$ |
|----------|-----------------------------------|------------------------------------|-------------------------------------|
| ZoomST   | <b>0.947</b><br>$\pm 0.019$       | <b>0.882</b><br>$\pm 0.046$        | <b>0.953</b><br>$\pm 0.021$         |
| CS       | 0.943<br>$\pm 0.019$              | 0.845<br>$\pm 0.044$               | 0.938<br>$\pm 0.019$                |
| No Cross | 0.933<br>$\pm 0.023$              | 0.827<br>$\pm 0.050$               | 0.933<br>$\pm 0.020$                |